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November 1, 1962

SUPERSEDING
Int. Fed. Spec. GG-C-00620b (GSA-FSS)
May 11, 1961

FEDERAL SPECIFICATION

COUNTER, RECIPROCATING, HAND-HELD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers hand-held reciprocating counters generally used for counting people, vehicles, trees, packages, blood cells, bacteriological colonies, and other visible objects.

1.1.1 *Federal specification coverage.* Federal specifications do not cover all types of the commodity indicated by the titles of the specification, but are intended to cover the types ordinarily used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.* Counters covered by this specification shall be of the following types and classes, as specified (see 6.1):

Type I.—Single operation zero reset (fig. 1).
Class 2.—Zero to 9999 minimum registering range.

Type II.—Multiple operation zero reset (fig. 2).
Class 1.—Zero to 999 minimum registering range.
Class 2.—Zero to 9999 minimum registering range.

NOTE: At the option of the supplier, type I counters may be bid in competition with any type II counter.

2. APPLICABLE SPECIFICATIONS, AND STANDARDS.

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-585—Boxes; Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box; Fiberboard.
PPP-T-76—Tape, Pressure Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in

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the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of counters which are otherwise in accordance with the requirements of this specification.

3.2 Description. The counter shall be of the wheel register type.

3.3 Material. Material shall be as specified herein. Material not definitely specified shall be of a good commercial quality, entirely suitable for the purpose intended. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.4 Marking. The counter shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.5 Construction. The counter shall consist essentially of a case, registering wheels, operating key, and either a holding ring or other means of holding firmly in the hand. The holding ring shall be attached to the case, shall fold flat against the case, and shall swing out to permit a finger to be inserted when the counter is operated by either hand. The numbers on the registering wheels, when assembled in the case and advanced by the operating key, shall stop in the center of the viewing window. All numerals shall remain in alignment within 1/32 inch until either advanced by the operating key or, when applicable, reset to the zero reading.

3.5.1 *Case.* The case shall be of corrosion-resistant material and shall not show evidence of breaking, chipping, or cracking when tested as specified in 4.4.1. The case shall be not more than 1⅛ inches thick and not more than 1⅞ inches in diameter. The case shall have a viewing window or windows made of clear glass or clear plastic through which the register numbers can be easily seen.

3.5.1.1 *Optional types of cases.* The case shall be of dust-free type except that, when specified (see 6.1), the case shall be of the dustproof type. The dust-free type of case shall be fastened together with screws or similar fasteners so as to facilitate examination of the internal mechanism. The dustproof type of case shall be sealed by riveting, welding, brazing, or similar method of fabricating to insure a dustproof enclosure.

3.5.2 *Registering wheels.* Each registering wheel shall be provided with ten numerals from 0 to 9 inclusive. The numerals shall be not less than ⅛-inch in height, well-proportioned and, for legibility, of a color contrasting to the background. Means shall be provided whereby the tens and hundreds

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units shall be advanced in proper sequence when the wheels are operated by the operating key.

3.5.3 *Operating key.* The operating key for advancing the numbers shall be either a metal plunger or lever projecting outside the case or a depressable cover of the case. Each depression of the key shall advance the register one consecutive number without skipping or advancing any number or numbers when tested as specified in 4.4.2.

3.5.4 *Holding ring or spring.* The holding ring shall be of wire at least 0.06 inch in diameter and the inside diameter of the ring shall be 1 inch plus or minus 3/16 inch. When specified (see 6.1), a spring of suitable tension and length may be used instead of a ring as a holding device.

3.6 Type I. The type I counter (fig. 1) shall be provided with a means whereby all registering wheels can be quickly and accurately returned to their zero reading. The resetting of all registering wheels to their zero position shall be accomplished by a single operation, such as a single turn of the reset knob. The reset knob shall have a knurled or serrated periphery for ease in turning.

3.6.1 *Class* 2. The type I, class 2 counter shall have a registering range from zero to 9999.

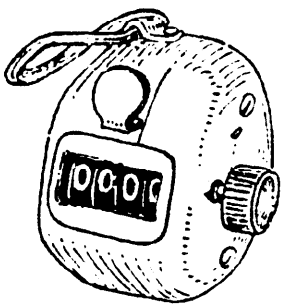


Figure 1. - Counter, reciprocating, hand-held. Type I, class 2 single operation zero reset.

3.7 Type II. The type II counter (fig. 2) shall be provided with means whereby all registering wheels can be quickly and accurately returned to their zero reading. The resetting of all registering wheels may be accomplished by single or multiple operation. Knob(s) shall be knurled or serrated on the periphery for ease in turning.

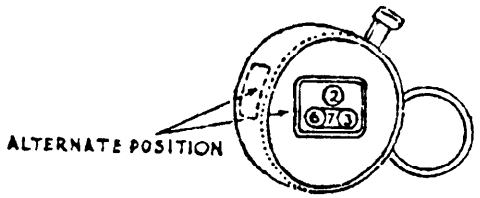


Figure 2. - Counter, reciprocating, hand-held. Type II, class 2, multiple operation zero reset.

3.7.1 *Class* 1. The type II, class 1 counter shall have a registering range from zero to at least 999. The counter shall provide for resetting all registering wheels to zero by employing not more than 12 operations, each operation consisting of any of the following: One depression of a key, one revolution of a reset knob, or any similar operation.

3.7.2 *Class* 2. The type II, class 2 counter shall have a registering range from zero to 9999. The counter shall provide for resetting all registering wheels to zero by employing not more than 25 operations.

3.8 Finish. Unless otherwise specified (see 6.1), the counter shall be provided with a perspiration-resistant finish in accordance with the manufacturer's standard practice.

3.9 Workmanship. Workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.* The inspection lot size shall consist of all units of product submitted for inspection at one time.

4.2 Inspection. Inspection shall consist of visual examinations and tests as indicated herein to assure compliance with specification requirements.

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4.2.1 *Contractor inspection.* The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities or services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government for a period of time as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.2.2 *Government verification.* Government verification will consist of surveillance of contractor's inspection system and inspection of his product to determine compliance with contract requirements.

4.3 **Sampling.** Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.3.1 *Visual examination.* Visual examination of counters shall be in accordance with the classification of defects furnished in table I. The following AQL's expressed in defects per 100 units shall apply:

Major 2.5
Total 10.0

TABLE I.—*Classification of defects*

Examine	Defect	Major	Minor
Case	Not made of corrosion-resisting material	X	
	More than 1-1/8 inches thick		X
	Manufacturer's name or trademark not legible or permanently marked		X
Registering wheel	Not provided with ten numerals from 0 to 9	X	
	Numerals less than 1/8 inch in height		X
	Not in contrasting color and clearly legible	X	
	Numerals not properly aligned	X	
	Tens, hundreds, and thousands units do not advance when preceding wheel moves from 9 to 0	X	
Operating key	Not specified type of key	X	
	Binds, sticks, or does not operate with ease	X	
Holding ring or spring	Ring wire less than 0.06 inches in diameter*		X
	Diameter of ring less than 13/16 or more than 1 and 3/16 inches*		X
Counter	Component loosely attached	X	
	Component missing	X	
	Component damaged affecting serviceability	X	
	Component damaged affecting appearance but not serviceability		X
Type I	Not provided with single operation for reset to zero	X	
	Range less than zero to 9999	X	
Type II, class 1	Reset knobs not knurled or serrated		X
	Range less than zero to 999	X	
	Resetting to zero requires more than 12 operations	X	
Type II, class 2	Range less than zero to 9999	X	
	Resetting to zero requires more than 25 operations	X	

* Measure 5 rings, check visually the balance of the rings in the sample size. When in doubt, measure all rings in sample size.

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4.3.2 *Testing.* Sampling for tests shall be in accordance with Inspection Level L-5. The AQL's shall be 6.5 percent defective. The sample unit shall be one complete counter. Any failure on the drop test, accuracy test, or count life test shall render the sample unit defective.

4.4 Tests. The samples shall be tested as specified in 4.4.1, 4.4.2, and 4.4.3.

4.4.1 *Drop test.* The complete counter shall be subjected to a drop test to determine the strength of the case and the working mechanism. The complete counter shall be dropped five times from a height of three feet onto a concrete floor without evidence of breaking, chipping, or cracking the case. Dents in the case will be permitted as long as the counter passes the accuracy test.

4.4.2 *Accuracy test.* The inspector shall operate the key on the counters tested in accordance with 4.4.1, so as to advance all registering wheels. This shall be done as fast as the inspector can operate the counter and keep track of the count with the viewing window covered, then uncover the window to verify that the tally is correct. The inspector may repeat this operation several times to insure that the counter does not skip or advance any numbers.

4.4.3 *Count life test.* Each sample counter shall be subjected to a count life test of 10 million counts, in accordance with the following method: (1) Secure the counter to a bench; (2) a reciprocating bar with rubber bumper attached shall actuate the operating key, or button, 10 million times at a continuous speed of 300 counts per minute.

NOTE: The bar and rubber bumper shall be adjusted to provide 3 to 5 ounces of pressure on the operating key or button after the key or button is fully depressed against a positive stop. After the count life test is completed the sample counter shall also meet the accuracy test method in 4.4.2.

5. PREPARATION FOR DELIVERY

(Civil agencies. Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of preservation, packaging, and packing).

5.1 Preservation and packaging.

5.1.1 *Level A.*

5.1.1.1 *Preservation and unit packaging.* The counters shall be individually preserved and unit packaged in conformance with MIL-P-116, method IA 8, without preservative, using paperboard folding cartons conforming to PPP-B-566, style, type, and class optional, as the containers. The dip-coat-sealed packages shall be over-wrapped with kraft paper suitably secured.

5.1.1.2 *Intermediate packaging.* Unit packages of counters shall be packaged together in quantities of 50 each, in snug-fitting fiberboard boxes conforming to PPP-B-636, compliance symbol W6s or W6c. Suitable cushioning shall be provided to prevent movement within the boxes. The boxes shall be closed and sealed in conformance with the appendix to the box specification, and in addition, all joints and seams, including those occurring in the manufacture of the boxes, shall be covered with minimum 3-inch wide gummed tape conforming to PPP-T-76, extended over all corners and edges at least 3 inches.

5.1.2 *Level C.*

5.1.2.1 *Preservation.* Preservation shall be in accordance with the manufacturer's standard practice.

5.1.2.2 *Packaging.* Commercial packaging will be acceptable.

5.2 Packing.

5.2.1 *Level A.* Intermediate packages of counters shall be packed together in quantities not to exceed 70 pounds gross weight in snug-fitting nailed wood boxes conforming to PPP-B-621, style 2, 2½, or 3, or wood-cleated plywood boxes conforming to PPP-B-601, style A or B, for type 2 load. Suit-

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able cushioning shall be provided to prevent movement within the boxes. The boxes shall be closed and strapped in conformance with the appendix to the applicable box specification.

5.2.2 *Level B.* The counters shall be packed in wood-cleated fiberboard, wood-cleated-plywood, nailed wood, wirebound wood, corrugated fiber, or solid fiber boxes conforming to PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-585, and PPP-B-636, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds. The gross weight of fiber-board boxes shall be subject to the limitation of the box specification.

5.2.3 *Level C.* The counters shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, in such a manner as to insure acceptance and safe delivery to the designated point by common or other carriers, at the lowest rate of the carrier.

5.3 **Marking of shipments.**

5.3.1 *Military.* In addition to any special marking required (see 6.1), unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 *Civil agencies.* In addition to any special marking required (see 6.1), unit packages, intermediate packages, and shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type and class required (see 1.2).

(c) If a sealed dustproof case is required, so state (see 3.5.1.1).

(d) Finish, if different (see 3.8).

(e) Level of packaging and packing required (see section 5).

(f) Special marking, if required (see 5.3.1 and 5.3.2).

6.2 **Illustrations.** The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of counters otherwise conforming to this specification.

6.3 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Numbering machines, not otherwise indexed by name.

Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Numbering machines, not otherwise indexed.

Motor volume minimum weight 24,000 pounds.

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